

# Student Attrition Risk Predictions

Sprint 2 machine-learning output

**Source of truth:** attached notebook Machine Learning Model.ipynb (Databricks export dated 2026-08-16) together with the approved SDD set after amendment A-6.

**Prepared:** 20 August 2026

## 1. Canonical output

| Property                                        | Approved value                                                                          |
|-------------------------------------------------|-----------------------------------------------------------------------------------------|
| Unity Catalog table                             | <code>workspace.student_aggregate.student_attrition_ris</code>                          |
| Grain                                           | One row per <code>student_deidentified_hash</code>                                      |
| Write mode                                      | Overwrite of the latest validated snapshot                                              |
| Inference population                            | Every valid student row in the Phase 1 canonical modelling projection                   |
| Population size (executed evidence, 2026-08-06) | 973,770 students / 973,770 rows                                                         |
| Persisted columns                               | Exactly six (unrounded probability is validated in staging, then dropped)               |
| Source tables modified                          | None                                                                                    |
| Transient staging table                         | <code>student_attrition_risk_prediction__scoring_staging</code> (dropped after publish) |

## 2. Persisted schema

| Column                                 | Type    | Purpose                                                                                                                    |
|----------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------|
| <code>student_deidentified_hash</code> | string  | Logical primary key used by the Databricks application to retrieve the student's result.                                   |
| <code>attrition_risk_percentage</code> | double  | Positive-class probability multiplied by 100. Range 0 to 100. Ranking score, not a calibrated likelihood (limitation L-3). |
| <code>attrition_risk_flag</code>       | boolean | True when unrounded probability $\geq 0.50$ ; false when $< 0.50$ . Never derived from the rounded percentage.             |

|                      |           |                                               |
|----------------------|-----------|-----------------------------------------------|
| prediction_threshold | double    | Stored value 0.50 for traceability.           |
| mlflow_run_id        | string    | MLflow run that produced the scored snapshot. |
| scored_at            | timestamp | UTC timestamp at which inference completed.   |

Intermediate column used only in staging (not persisted): `attrition_risk_probability` = Spark ML probability vector slot 1 (`vector_to_array(probability)[1]`).

### 3. How each prediction is produced (current notebook)

1. Phase 1 builds one modelling row per student from the fact table plus the course and teaching-period dimensions.
2. Phase 2 derives numeric medians and categorical levels from the training split only, applies deterministic manual one-hot encoding with an unseen/NULL bucket, and assembles the vector with `VectorAssembler.transform()` only.
3. `RandomForestClassifier` is fitted directly to the prepared training vector (100 trees, max depth 7, min instances per node 20, feature subset sqrt, seed 42).
4. Phase 3 reuses the same `prepare_feature_vector` contract. Preprocessing is not refitted.
5. The fitted forest scores every applicable student once. The result is written to the transient staging table, validated, then published as the six-column snapshot.

```
attrition_risk_probability = vector_to_array(__probability)[1] attrition_risk_percentage =
attrition_risk_probability * 100 attrition_risk_flag = attrition_risk_probability >= 0.50
```

Therefore 0.50 is flagged true and 0.49 is false. Rounding is not applied before the flag.

### 4. Inference population

Clarified Sprint 2 decision: score every valid student row in the prepared modelling dataset, not only the test split.

| Group                             | Meaning                                                                                                                                                                                                |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Training / validation / test rows | All scored so the application has a current result for every student. Training rows are scored optimistically; this affects stored percentages, not Phase 2 metrics (measured on untouched test data). |
| Rows with unknown outcome         | Excluded from supervised training, but scored because they are exactly the students the application needs a prediction for.                                                                            |

Executed Phase 1 evidence (2026-08-06): 973,770 fact rows, 973,770 distinct students, 0 duplicate students, 0 NULL identifiers, 0 NULL targets. Target base rate 53,694 positive (5.51%) and 920,076 negative (94.49%).

### 5. Pre-publish validation (T27)

Nothing is written to the approved table until all of the following pass against the staging snapshot:

- one output row per applicable student;
- no null and no duplicate student identifiers;

- no null intermediate probabilities; probabilities in [0, 1];
- percentages in [0, 100];
- Boolean flags only;
- threshold stored as exactly 0.50;
- MLflow run ID populated on every row;
- flag matches the unrounded threshold rule row by row;
- output count matches the Phase 1 inference population.

Post-publish checks query the approved Delta table itself, so they verify what the application will read.

## 6. How to retrieve live predictions

```
SELECT student_deidentified_hash, attrition_risk_percentage, attrition_risk_flag,
prediction_threshold, mlflow_run_id, scored_at FROM
workspace.student_aggregate.student_attrition_risk_prediction WHERE student_deidentified_hash =
'<hash>';
```

```
SELECT COUNT(*) AS rows, COUNT(DISTINCT student_deidentified_hash) AS students, SUM(CASE WHEN
attrition_risk_flag THEN 1 ELSE 0 END) AS flagged_true, MIN(attrition_risk_percentage) AS
min_pct, AVG(attrition_risk_percentage) AS mean_pct, MAX(attrition_risk_percentage) AS max_pct
FROM workspace.student_aggregate.student_attrition_risk_prediction;
```

## 7. How stored percentages must be read

Limitation L-3 (approved at Phase 1 and accepted at Phase 3): class weighting recentres predicted probabilities near 0.50. A stored value of 52 does not mean a 52% chance of leaving when the synthetic base rate is about 5.5%.

`attrition_risk_percentage` is a relative ranking score for advisor support, not a calibrated probability and not an automatic intervention decision.

Limitations L-1 and L-2 further constrain discrimination on this synthetic dataset (field-of-education decoupling; features nearly flat against the target). Low ROC-AUC near 0.5 is the expected result for this data, not a reason to change the approved features, parameters, or threshold.

## 8. Traceability to implementation

| Notebook / task | Output behaviour                                                                           |
|-----------------|--------------------------------------------------------------------------------------------|
| Phase 3 T22     | Build inference dataset from Phase 1 projection + training-derived preprocessing contract. |
| Phase 3 T23     | Apply contract and fitted Random Forest to every valid student row.                        |
| Phase 3 T24     | Extract positive probability; compute percentage = probability * 100.                      |
| Phase 3 T25     | Flag = probability >= 0.50.                                                                |
| Phase 3 T26-T28 | Stage, validate, overwrite approved six-column Delta table.                                |

|             |                                                                                    |
|-------------|------------------------------------------------------------------------------------|
| Phase 3 T29 | Retrieve by <code>student_deidentified_hash</code> and confirm application lookup. |
|-------------|------------------------------------------------------------------------------------|